

10.5 DMA registers

The DMA registers have to be accessed by words (32 bits).

10.5.1 DMA low interrupt status register (DMA_LISR)

Address offset: 0x00

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				TCIF3	HTIF3	TEIF3	DMEIF3	Reserv ed	FEIF3	TCIF2	HTIF2	TEIF2	DMEIF2	Reserv ed	FEIF2
r	r	r	r	r	r	r	r		r	r	r	r	r		r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				TCIF1	HTIF1	TEIF1	DMEIF1	Reserv ed	FEIF1	TCIF0	HTIF0	TEIF0	DMEIF0	Reserv ed	FEIF0
r	r	r	r	r	r	r	r		r	r	r	r	r		r

Bits 31:28, 15:12 Reserved, must be kept at reset value.

Bits 27, 21, 11, 5 **TCIFx**: Stream x transfer complete interrupt flag (x = 3..0)

This bit is set by hardware. It is cleared by software writing 1 to the corresponding bit in the DMA_LIFCR register.

0: No transfer complete event on stream x

1: A transfer complete event occurred on stream x

Bits 26, 20, 10, 4 **HTIFx**: Stream x half transfer interrupt flag (x=3..0)

This bit is set by hardware. It is cleared by software writing 1 to the corresponding bit in the DMA_LIFCR register.

0: No half transfer event on stream x

1: A half transfer event occurred on stream x

Bits 25, 19, 9, 3 **TEIFx**: Stream x transfer error interrupt flag (x=3..0)

This bit is set by hardware. It is cleared by software writing 1 to the corresponding bit in the DMA_LIFCR register.

0: No transfer error on stream x

1: A transfer error occurred on stream x

Bits 24, 18, 8, 2 **DMEIFx**: Stream x direct mode error interrupt flag (x=3..0)

This bit is set by hardware. It is cleared by software writing 1 to the corresponding bit in the DMA_LIFCR register.

0: No Direct Mode Error on stream x

1: A Direct Mode Error occurred on stream x

Bits 23, 17, 7, 1 Reserved, must be kept at reset value.

Bits 22, 16, 6, 0 **FEIFx**: Stream x FIFO error interrupt flag (x=3..0)

This bit is set by hardware. It is cleared by software writing 1 to the corresponding bit in the DMA_LIFCR register.

0: No FIFO Error event on stream x

1: A FIFO Error event occurred on stream x

10.5.2 DMA high interrupt status register (DMA_HISR)

Address offset: 0x04

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				TCIF7	HTIF7	TEIF7	DMEIF7	Reserv ed	FEIF7	TCIF6	HTIF6	TEIF6	DMEIF6	Reserv ed	FEIF6
				r	r	r	r		r	r	r	r	r		r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				TCIF5	HTIF5	TEIF5	DMEIF5	Reserv ed	FEIF5	TCIF4	HTIF4	TEIF4	DMEIF4	Reserv ed	FEIF4
				r	r	r	r		r	r	r	r	r		r

Bits 31:28, 15:12 Reserved, must be kept at reset value.

Bits 27, 21, 11, 5 **TCIFx**: Stream x transfer complete interrupt flag (x=7..4)

This bit is set by hardware. It is cleared by software writing 1 to the corresponding bit in the DMA_HIFCR register.

0: No transfer complete event on stream x

1: A transfer complete event occurred on stream x

Bits 26, 20, 10, 4 **HTIFx**: Stream x half transfer interrupt flag (x=7..4)

This bit is set by hardware. It is cleared by software writing 1 to the corresponding bit in the DMA_HIFCR register.

0: No half transfer event on stream x

1: A half transfer event occurred on stream x

Bits 25, 19, 9, 3 **TEIFx**: Stream x transfer error interrupt flag (x=7..4)

This bit is set by hardware. It is cleared by software writing 1 to the corresponding bit in the DMA_HIFCR register.

0: No transfer error on stream x

1: A transfer error occurred on stream x

Bits 24, 18, 8, 2 **DMEIFx**: Stream x direct mode error interrupt flag (x=7..4)

This bit is set by hardware. It is cleared by software writing 1 to the corresponding bit in the DMA_HIFCR register.

0: No Direct mode error on stream x

1: A Direct mode error occurred on stream x

Bits 23, 17, 7, 1 Reserved, must be kept at reset value.

Bits 22, 16, 6, 0 **FEIFx**: Stream x FIFO error interrupt flag (x=7..4)

This bit is set by hardware. It is cleared by software writing 1 to the corresponding bit in the DMA_HIFCR register.

0: No FIFO error event on stream x

1: A FIFO error event occurred on stream x

10.5.3 DMA low interrupt flag clear register (DMA_LIFCR)

Address offset: 0x08

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				CTCIF3	CHTIF3	CTEIF3	CDMEIF3	Reserved	CFEIF3	CTCIF2	CHTIF2	CTEIF2	CDMEIF2	Reserved	CFEIF2
				w	w	w	w		w	w	w	w	w		w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				CTCIF1	CHTIF1	CTEIF1	CDMEIF1	Reserved	CFEIF1	CTCIF0	CHTIF0	CTEIF0	CDMEIF0	Reserved	CFEIF0
				w	w	w	w		w	w	w	w	w		w

Bits 31:28, 15:12 Reserved, must be kept at reset value.

Bits 27, 21, 11, 5 **CTCIFx**: Stream x clear transfer complete interrupt flag (x = 3..0)

Writing 1 to this bit clears the corresponding TCIFx flag in the DMA_LISR register

Bits 26, 20, 10, 4 **CHTIFx**: Stream x clear half transfer interrupt flag (x = 3..0)

Writing 1 to this bit clears the corresponding HTIFx flag in the DMA_LISR register

Bits 25, 19, 9, 3 **CTEIFx**: Stream x clear transfer error interrupt flag (x = 3..0)

Writing 1 to this bit clears the corresponding TEIFx flag in the DMA_LISR register

Bits 24, 18, 8, 2 **CDMEIFx**: Stream x clear direct mode error interrupt flag (x = 3..0)

Writing 1 to this bit clears the corresponding DMEIFx flag in the DMA_LISR register

Bits 23, 17, 7, 1 Reserved, must be kept at reset value.

Bits 22, 16, 6, 0 **CFEIFx**: Stream x clear FIFO error interrupt flag (x = 3..0)

Writing 1 to this bit clears the corresponding CFEIFx flag in the DMA_LISR register

10.5.4 DMA high interrupt flag clear register (DMA_HIFCR)

Address offset: 0x0C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				CTCIF7	CHTIF7	CTEIF7	CDMEIF7	Reserved	CFEIF7	CTCIF6	CHTIF6	CTEIF6	CDMEIF6	Reserved	CFEIF6
				w	w	w	w		w	w	w	w	w		w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved				CTCIF5	CHTIF5	CTEIF5	CDMEIF5	Reserved	CFEIF5	CTCIF4	CHTIF4	CTEIF4	CDMEIF4	Reserved	CFEIF4
				w	w	w	w		w	w	w	w	w		w

Bits 31:28, 15:12 Reserved, must be kept at reset value.

Bits 27, 21, 11, 5 **CTCIFx**: Stream x clear transfer complete interrupt flag (x = 7..4)

Writing 1 to this bit clears the corresponding TCIFx flag in the DMA_HISR register

Bits 26, 20, 10, 4 **CHTIFx**: Stream x clear half transfer interrupt flag (x = 7..4)

Writing 1 to this bit clears the corresponding HTIFx flag in the DMA_HISR register

Bits 25, 19, 9, 3 **CTEIFx**: Stream x clear transfer error interrupt flag (x = 7..4)

Writing 1 to this bit clears the corresponding TEIFx flag in the DMA_HISR register

Bits 24, 18, 8, 2 **CDMEIFx**: Stream x clear direct mode error interrupt flag (x = 7..4)

Writing 1 to this bit clears the corresponding DMEIFx flag in the DMA_HISR register

Bits 23, 17, 7, 1 Reserved, must be kept at reset value.

Bits 22, 16, 6, 0 **CFEIFx**: Stream x clear FIFO error interrupt flag (x = 7..4)

Writing 1 to this bit clears the corresponding CFEIFx flag in the DMA_HISR register

10.5.5 DMA stream x configuration register (DMA_SxCR) (x = 0..7)

This register is used to configure the concerned stream.

Address offset: $0x10 + 0x18 \times \text{stream number}$

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				CHSEL[2:0]			MBURST [1:0]		PBURST[1:0]		Reserved	CT	DBM	PL[1:0]	
				rw	rw	rw	rw	rw	rw	rw		rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PINCOS	MSIZE[1:0]		PSIZE[1:0]		MINC	PINC	CIRC	DIR[1:0]		PFCTRL	TCIE	HTIE	TEIE	DMEIE	EN
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:28 Reserved, must be kept at reset value.

Bits 27:25 **CHSEL[2:0]**: Channel selection

These bits are set and cleared by software.

000: channel 0 selected

001: channel 1 selected

010: channel 2 selected

011: channel 3 selected

100: channel 4 selected

101: channel 5 selected

110: channel 6 selected

111: channel 7 selected

These bits are protected and can be written only if EN is '0'

Bits 24:23 **MBURST**: Memory burst transfer configuration

These bits are set and cleared by software.

00: single transfer

01: INCR4 (incremental burst of 4 beats)

10: INCR8 (incremental burst of 8 beats)

11: INCR16 (incremental burst of 16 beats)

These bits are protected and can be written only if EN is '0'

In direct mode, these bits are forced to 0x0 by hardware as soon as bit EN= '1'.

Bits 22:21 **PBURST[1:0]**: Peripheral burst transfer configuration

These bits are set and cleared by software.

00: single transfer

01: INCR4 (incremental burst of 4 beats)

10: INCR8 (incremental burst of 8 beats)

11: INCR16 (incremental burst of 16 beats)

These bits are protected and can be written only if EN is '0'

In direct mode, these bits are forced to 0x0 by hardware.

Bit 20 Reserved, must be kept at reset value.

Bit 19 **CT**: Current target (only in double buffer mode)

This bit is set and cleared by hardware. It can also be written by software.

0: The current target memory is Memory 0 (addressed by the DMA_SxM0AR pointer)

1: The current target memory is Memory 1 (addressed by the DMA_SxM1AR pointer)

This bit can be written only if EN is '0' to indicate the target memory area of the first transfer.

Once the stream is enabled, this bit operates as a status flag indicating which memory area is the current target.

Bit 18 **DBM**: Double buffer mode

This bit is set and cleared by software.

0: No buffer switching at the end of transfer

1: Memory target switched at the end of the DMA transfer

This bit is protected and can be written only if EN is '0'.

Bits 17:16 **PL[1:0]**: Priority level

These bits are set and cleared by software.

00: Low

01: Medium

10: High

11: Very high

These bits are protected and can be written only if EN is '0'.

Bit 15 **PINCOS**: Peripheral increment offset size

This bit is set and cleared by software

0: The offset size for the peripheral address calculation is linked to the PSIZE

1: The offset size for the peripheral address calculation is fixed to 4 (32-bit alignment).

This bit has no meaning if bit PINC = '0'.

This bit is protected and can be written only if EN = '0'.

This bit is forced low by hardware when the stream is enabled (bit EN = '1') if the direct mode is selected or if PBURST are different from "00".

Bits 14:13 **MSIZE[1:0]**: Memory data size

These bits are set and cleared by software.

00: byte (8-bit)

01: half-word (16-bit)

10: word (32-bit)

11: reserved

These bits are protected and can be written only if EN is '0'.

In direct mode, MSIZE is forced by hardware to the same value as PSIZE as soon as bit EN = '1'.

Bits 12:11 **PSIZE[1:0]**: Peripheral data size

These bits are set and cleared by software.

00: Byte (8-bit)

01: Half-word (16-bit)

10: Word (32-bit)

11: reserved

These bits are protected and can be written only if EN is '0'.

Bit 10 **MINC**: Memory increment mode

This bit is set and cleared by software.

0: Memory address pointer is fixed

1: Memory address pointer is incremented after each data transfer (increment is done according to MSIZE)

This bit is protected and can be written only if EN is '0'.

Bit 9 PINC: Peripheral increment mode

This bit is set and cleared by software.

0: Peripheral address pointer is fixed

1: Peripheral address pointer is incremented after each data transfer (increment is done according to PSIZE)

This bit is protected and can be written only if EN is '0'.

Bit 8 CIRC: Circular mode

This bit is set and cleared by software and can be cleared by hardware.

0: Circular mode disabled

1: Circular mode enabled

When the peripheral is the flow controller (bit PFCTRL=1) and the stream is enabled (bit EN=1), then this bit is automatically forced by hardware to 0.

It is automatically forced by hardware to 1 if the DBM bit is set, as soon as the stream is enabled (bit EN = '1').

Bits 7:6 DIR[1:0]: Data transfer direction

These bits are set and cleared by software.

00: Peripheral-to-memory

01: Memory-to-peripheral

10: Memory-to-memory

11: reserved

These bits are protected and can be written only if EN is '0'.

Bit 5 PFCTRL: Peripheral flow controller

This bit is set and cleared by software.

0: The DMA is the flow controller

1: The peripheral is the flow controller

This bit is protected and can be written only if EN is '0'.

When the memory-to-memory mode is selected (bits DIR[1:0]=10), then this bit is automatically forced to 0 by hardware.

Bit 4 TCIE: Transfer complete interrupt enable

This bit is set and cleared by software.

0: TC interrupt disabled

1: TC interrupt enabled

Bit 3 HTIE: Half transfer interrupt enable

This bit is set and cleared by software.

0: HT interrupt disabled

1: HT interrupt enabled

Bit 2 TEIE: Transfer error interrupt enable

This bit is set and cleared by software.

0: TE interrupt disabled

1: TE interrupt enabled

Bit 1 DMEIE: Direct mode error interrupt enable

This bit is set and cleared by software.

0: DME interrupt disabled

1: DME interrupt enabled

Bit 0 **EN**: Stream enable / flag stream ready when read low

This bit is set and cleared by software.

0: Stream disabled

1: Stream enabled

This bit may be cleared by hardware:

- on a DMA end of transfer (stream ready to be configured)
- if a transfer error occurs on the AHB master buses
- when the FIFO threshold on memory AHB port is not compatible with the size of the burst

When this bit is read as 0, the software is allowed to program the Configuration and FIFO bits registers. It is forbidden to write these registers when the EN bit is read as 1.

Note: Before setting EN bit to '1' to start a new transfer, the event flags corresponding to the stream in DMA_LISR or DMA_HISR register must be cleared.

10.5.6 DMA stream x number of data register (DMA_SxNDTR) (x = 0..7)

Address offset: $0x14 + 0x18 \times \text{stream number}$

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NDT[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:16 Reserved, must be kept at reset value.

Bits 15:0 **NDT[15:0]**: Number of data items to transfer

Number of data items to be transferred (0 up to 65535). This register can be written only when the stream is disabled. When the stream is enabled, this register is read-only, indicating the remaining data items to be transmitted. This register decrements after each DMA transfer.

Once the transfer has completed, this register can either stay at zero (when the stream is in normal mode) or be reloaded automatically with the previously programmed value in the following cases:

- when the stream is configured in Circular mode.
- when the stream is enabled again by setting EN bit to '1'

If the value of this register is zero, no transaction can be served even if the stream is enabled.

10.5.7 DMA stream x peripheral address register (DMA_SxPAR) (x = 0..7)

Address offset: $0x18 + 0x18 \times \text{stream number}$

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
PAR[31:16]															
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PAR[15:0]															
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:0 **PAR[31:0]**: Peripheral address

Base address of the peripheral data register from/to which the data are read/written.

These bits are write-protected and can be written only when bit EN = '0' in the DMA_SxCR register.

10.5.8 DMA stream x memory 0 address register (DMA_SxM0AR) (x = 0..7)

Address offset: $0x1C + 0x18 \times \text{stream number}$

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
M0A[31:16]															
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
M0A[15:0]															
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:0 **M0A[31:0]**: Memory 0 address

Base address of Memory area 0 from/to which the data are read/written.

These bits are write-protected. They can be written only if:

- the stream is disabled (bit EN = '0' in the DMA_SxCR register) or
- the stream is enabled (EN = '1' in DMA_SxCR register) and bit CT = '1' in the DMA_SxCR register (in Double buffer mode).

10.5.9 DMA stream x memory 1 address register (DMA_SxM1AR) (x = 0..7)

Address offset: $0x20 + 0x18 \times \text{stream number}$

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
M1A[31:16]															
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
M1A[15:0]															
rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW	rW

Bits 31:0 **M1A[31:0]**: Memory 1 address (used in case of Double buffer mode)

Base address of Memory area 1 from/to which the data are read/written.

This register is used only for the Double buffer mode.

These bits are write-protected. They can be written only if:

- the stream is disabled (bit EN= '0' in the DMA_SxCR register) or
- the stream is enabled (EN='1' in DMA_SxCR register) and bit CT = '0' in the DMA_SxCR register.

10.5.10 DMA stream x FIFO control register (DMA_SxFCR) (x = 0..7)

Address offset: $0x24 + 0x18 \times \text{stream number}$

Reset value: 0x0000 0021

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved								FEIE	Reserved	FS[2:0]			DMDIS	FTH[1:0]	
								rw		r	r	r	rw	rw	rw

Bits 31:8 Reserved, must be kept at reset value.

Bit 7 **FEIE**: FIFO error interrupt enable

This bit is set and cleared by software.

0: FE interrupt disabled

1: FE interrupt enabled

Bit 6 Reserved, must be kept at reset value.

Bits 5:3 **FS[2:0]**: FIFO status

These bits are read-only.

000: $0 < \text{fifo_level} < 1/4$

001: $1/4 \leq \text{fifo_level} < 1/2$

010: $1/2 \leq \text{fifo_level} < 3/4$

011: $3/4 \leq \text{fifo_level} < \text{full}$

100: FIFO is empty

101: FIFO is full

others: no meaning

These bits are not relevant in the direct mode (DMDIS bit is zero).

Bit 2 **DMDIS**: Direct mode disable

This bit is set and cleared by software. It can be set by hardware.

0: Direct mode enabled

1: Direct mode disabled

This bit is protected and can be written only if EN is '0'.

This bit is set by hardware if the memory-to-memory mode is selected (DIR bit in DMA_SxCR are "10") and the EN bit in the DMA_SxCR register is '1' because the direct mode is not allowed in the memory-to-memory configuration.

Bits 1:0 **FTH[1:0]**: FIFO threshold selection

These bits are set and cleared by software.

00: 1/4 full FIFO

01: 1/2 full FIFO

10: 3/4 full FIFO

11: full FIFO

These bits are not used in the direct mode when the DMIS value is zero.

These bits are protected and can be written only if EN is '0'.

Table 52 summarizes the DMA registers.

[illegible]

Table 52. DMA register map and reset values (continued)[illegible]

Table 52. DMA register map and reset values (continued)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x0068	DMA_S3M1AR	M1A[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x006C	DMA_S3FCR	Reserved																								FEIE	Reserved	FS[2:0]		DMDIS	FTH [1:0]		
	Reset value																									0		1	0	0	0	0	0
0x0070	DMA_S4CR	Reserved				CHSEL[2:0]		MBURST[1:0]		PBURST[1:0]		Reserved	CT	DBM	PL[1:0]	PINCOS	MSIZE[1:0]	PSIZE[1:0]	MINC	PINC	CIRC	DIR [1:0]	PFCTRL	TCIE	HTIE	TEIE	DMEIE	EN					
	0					0	0	0	0	0	0																		0	0	0	0	0
0x0074	DMA_S4NDTR	Reserved															NDT[15:]																
	Reset value																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0078	DMA_S4PAR	PA[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x007C	DMA_S4M0AR	M0A[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0080	DMA_S4M1AR	M1A[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0084	DMA_S4FCR	Reserved																								FEIE	Reserved	FS[2:0]		DMDIS	FTH [1:0]		
	Reset value																									0		1	0	0	0	0	0
0x0088	DMA_S5CR	Reserved				CHSEL[2:0]		MBURST[1:0]		PBURST[1:0]		Reserved	CT	DBM	PL[1:0]	PINCOS	MSIZE[1:0]	PSIZE[1:0]	MINC	PINC	CIRC	DIR[1:0]	PFCTRL	TCIE	HTIE	TEIE	DMEIE	EN					
	0					0	0	0	0	0	0																		0	0	0	0	0
0x008C	DMA_S5NDTR	Reserved															NDT[15:]																
	Reset value																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0090	DMA_S5PAR	PA[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0094	DMA_S5M0AR	M0A[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x0098	DMA_S5M1AR	M1A[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x009C	DMA_S5FCR	Reserved																								FEIE	Reserved	FS[2:0]		DMDIS	FTH [1:0]		
	Reset value																									0		1	0	0	0	0	0
0x00A0	DMA_S6CR	Reserved				CHSEL[2:0]		MBURST[1:0]		PBURST[1:0]		Reserved	CT	DBM	PL[1:0]	PINCOS	MSIZE[1:0]	PSIZE[1:0]	MINC	PINC	CIRC	DIR[1:0]	PFCTRL	TCIE	HTIE	TEIE	DMEIE	EN					
	0					0	0	0	0	0	0																		0	0	0	0	0
0x00A4	DMA_S6NDTR	Reserved															NDT[15:]																
	Reset value																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x00A8	DMA_S6PAR	PA[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 52. DMA register map and reset values (continued)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x00AC	DMA_S6M0AR	M0A[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x00B0	DMA_S6M1AR	M1A[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x00B4	DMA_S6FCR	Reserved																								FEIE	Reserved	FS[2:0]		DMDIS		FTH [1:0]	
	Reset value																									0			1	0	0	0	0
0x00B8	DMA_S7CR	Reserved				CHSEL[2:0]		MBURST[1:0]		PBURST[1:0]		Reserved	CT	DBM	PL[1:0]		PINCOS	MSIZE[1:0]		PSIZE[1:0]		MINC	PINC	CIRC	DIR[1:0]		PFCTRL	TCIE	HTIE	TEIE	DMEIE	EN	
	Reset value												0	0	0	0	0	0	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x00BC	DMA_S7NDTR	Reserved															NDT[15:]																
	Reset value																0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x00C0	DMA_S7PAR	PA[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x00C4	DMA_S7M0AR	M0A[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x00C8	DMA_S7M1AR	M1A[31:0]																															
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x00CC	DMA_S7FCR	Reserved																								FEIE	Reserved	FS[2:0]		DMDIS		FTH [1:0]	
	Reset value																									0			1	0	0	0	0

Refer to [Section 2.3: Memory map](#) for the register boundary addresses.